

Innate Immunity I and II

Tony A. Slieman, Ph.D., M.Sc.

Director of Academic Affairs and Associate Professor

Founding Director of Master's in Biomedical Sciences

Department of Biomedical and Anatomical Sciences

tslieman@nyit.edu



Session Objectives

1. Compare and contrast innate and adaptive IR
2. Classify the innate IR into 1st and 2nd line
3. Distinguish the different components of the 1st and 2nd line of defense
4. Explain key innate processes of inflammation, phagocytosis, fever, interferon & complement action
 - a) Identify the role of cytokines in these processes
 - b) Identify the role of PRRs (pattern recognition receptors) in the inflammatory response
5. Identify deficiencies in innate immunity and the diseases it causes

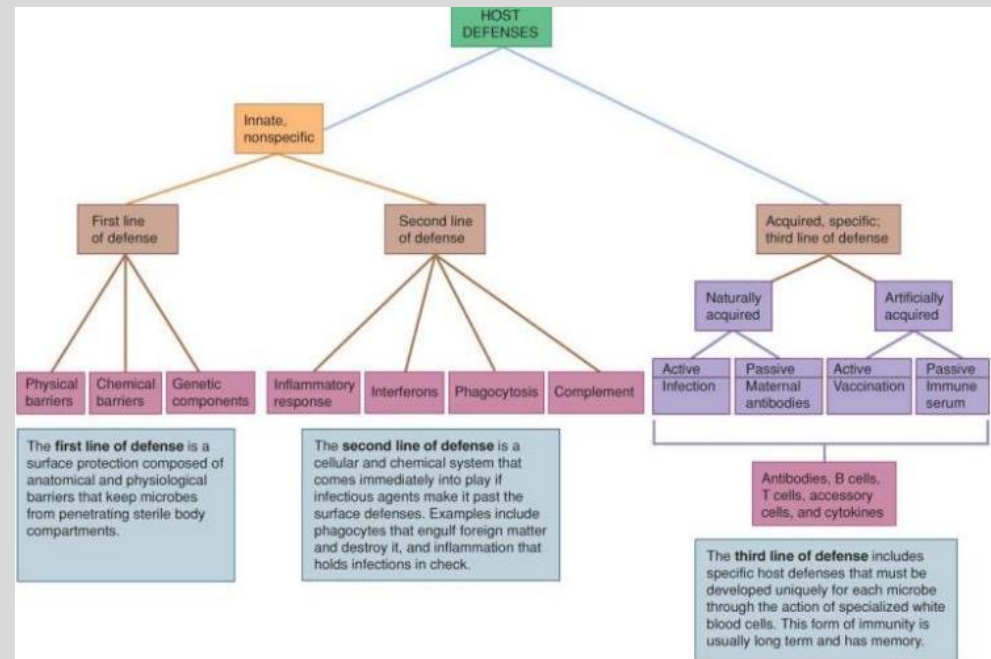
Innate and Adaptive IR

- Distinction between adaptive and innate is essentially artificial
- Some cells overlap/essential to both responses (e.g., Macrophages)
- Innate/adaptive interact

Innate IR	Adaptive IR
Non-specific	Specific
Doesn't learn	Learns (memory)
Fast response	Slow response

Innate IR

- First line and second line
- 1st line:
 1. Physical, chemical & biological
 2. skin, ear wax, defensins, tears, mucus, stomach acid, normal flora, bile, etc.
- 2nd line: inflammation, fever, complement, interferon (IFN), phagocytosis, & NK cells



First line: Physical

- Physical barriers:
 1. Skin
 2. Mucous membranes
 3. Respiratory track (*cilia, hair, mucous*)
 4. Urinary track (*voiding*)

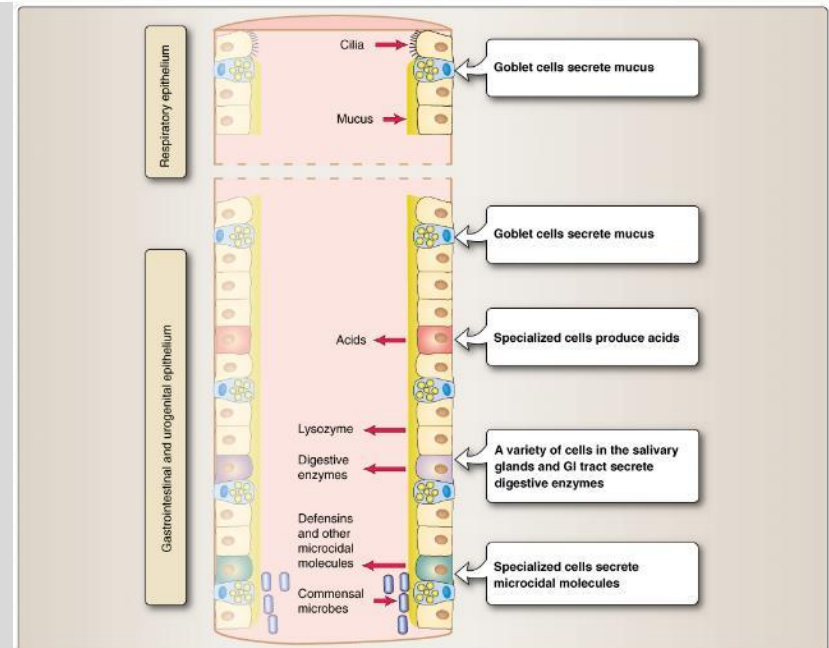
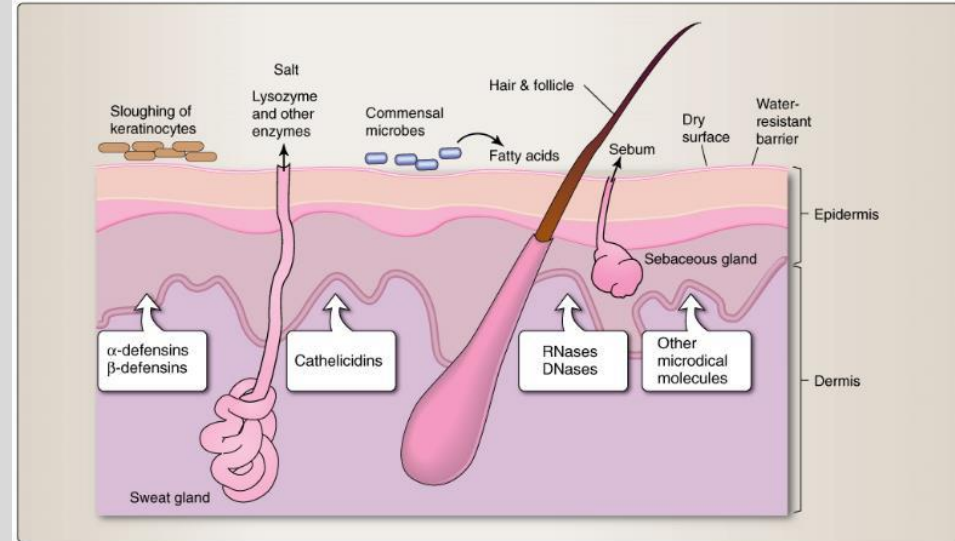
First line: Chemical

- Chemical barriers:

1. pH (skin = 5.5; stomach = 1-3; vagina = 4.5)

2. Microcidal molecules:

- a) Lysozyme
- b) Defensins
- c) Cathelicidins
- d) RNases/DNases
- e) Salt



First line: biological

Body Area		Common Organisms (Bacteria unless Otherwise Noted)
Skin		<i>Acinetobacter spp.</i> <i>Staphylococcus spp.</i>
	Scalp	<i>Malassezia spp.</i> (fungus)
	Oil glands	<i>Propionibacterium spp.</i>
Mouth/throat		<i>Actinomyces spp.</i> <i>Fusobacterium spp.</i> <i>Lactobacillus spp.</i> <i>Leptotrichia spp.</i> <i>Mycoplasma spp.</i> <i>Neisseria spp.</i> <i>Staphylococcus spp.</i> <i>Streptococcus spp.</i>
Nasal cavity/pharynx		<i>Corynebacterium spp.</i> <i>Haemophilus influenzae</i> <i>Neisseria meningitidis</i> <i>Staphylococcus spp.</i> <i>Streptococcus spp.</i>
Stomach		<i>Helicobacter pylori</i>
Small/large intestine		<i>Bacteroides spp.</i> <i>Bifidobacterium spp.</i> [in breast-fed infants] <i>Candida albicans</i> (fungus) <i>Clostridium spp.</i> <i>Enterobacter spp.</i> <i>Escherichia coli</i> <i>Klebsiella spp.</i> <i>Lactobacillus spp.</i> [in bottle-fed infants] <i>Proteus spp.</i> <i>Pseudomonas aeruginosa</i> <i>Streptococcus spp.</i>
Upper respiratory tract		<i>Corynebacterium catarrhalis</i> <i>Neisseria meningitidis</i> <i>Streptococcus spp.</i> (α-hemolytic)
Urogenital tract	Urethral opening	<i>Corynebacterium spp.</i> <i>Enterococcus faecalis</i> <i>Staphylococcus epidermidis</i>
	Vagina	<i>Candida albicans</i> (fungus) <i>Corynebacterium spp.</i> <i>Lactobacillus spp.</i> <i>Streptococcus spp.</i>
Eye	Surface	<i>Staphylococcus spp.</i> <i>Streptococcus spp.</i> <i>Branhamella catarrhalis</i>

Which of the following is an example of a normal physiological pH barrier to microbial colonization?

Respiratory track pH between 1 and 3

Skin pH of 5

Stomach pH between 9 and 11

Vaginal pH of 7

Second line

1. Inflammation
2. Fever
3. Interferon (IFN)
4. Complement
5. Phagocytosis
6. NK cell responses

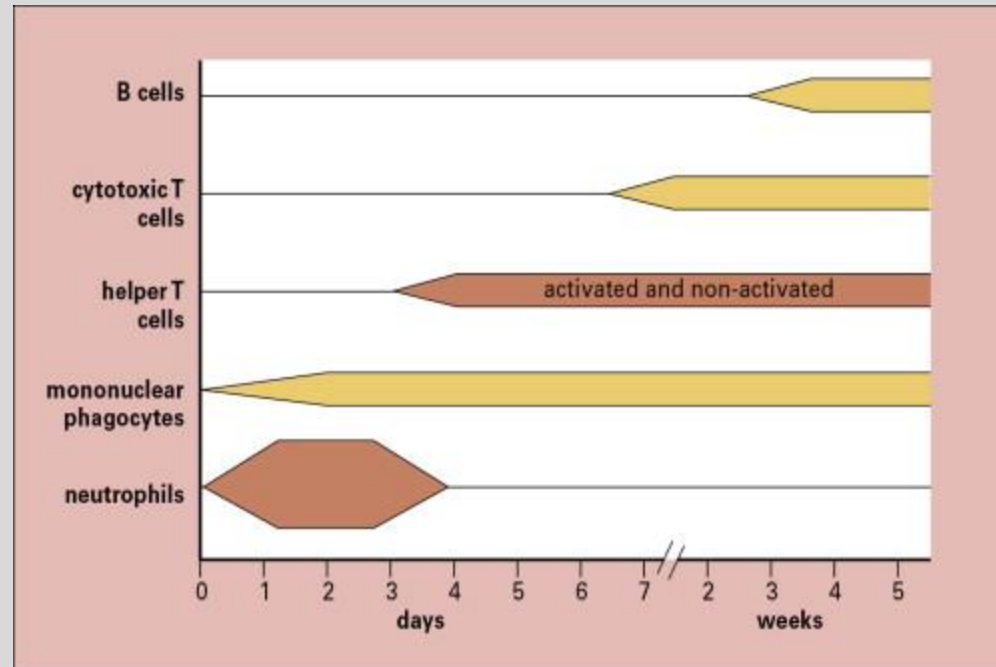
Second line: Inflammation

- A response to tissue damage (physical, infectious, etc.)
- Inflammatory mediators i.e., histamine, C5a, TNF- α
- Cardinal signs: Heat, Pain, Redness, Swelling, (From Latin: *Calor*, *dolor*, *rubor*, and *tumor*), & Loss of Function
- Coordinated by Macrophages



Second line: Inflammation

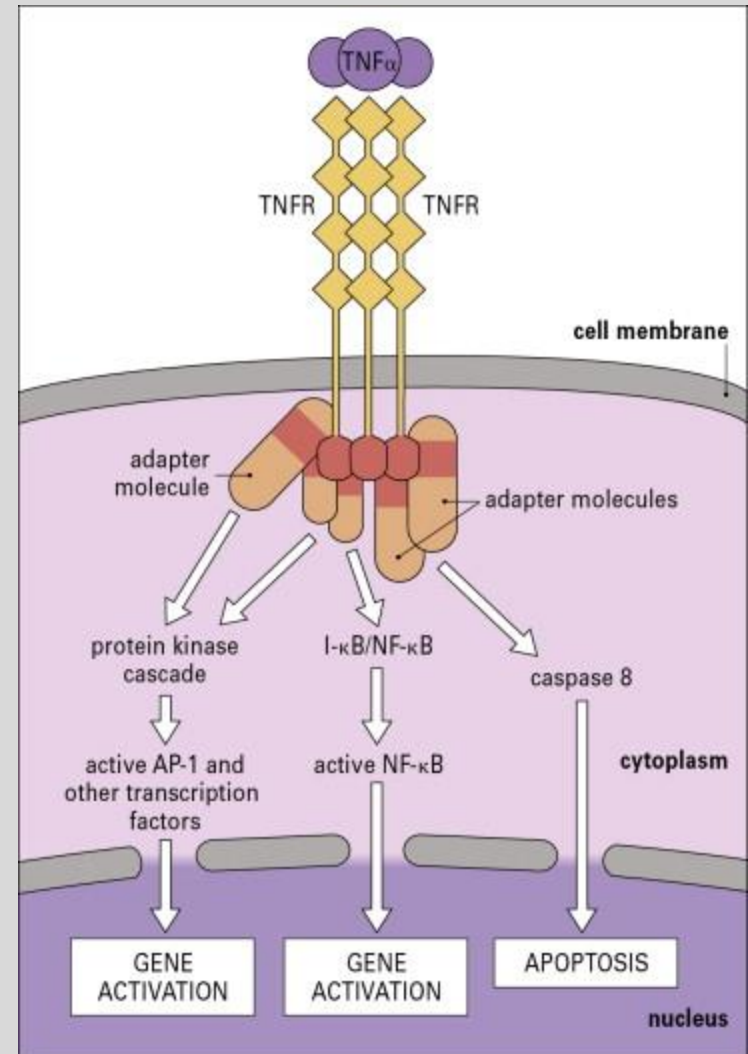
- Increase blood supply to area
- Increase in capillary permeability
- Increase in leukocyte chemotaxis
- Fast clearance of Ag → acute
- Persistent Ag → chronic



David Male, Jonathan Brostoff, David Roth, and Ivan Roitt. 2013. *Immunology* 8th edition. Elsevier publisher

Second line: Inflammation

- Cytokines: $\text{TNF-}\alpha$, IL-1 & $\text{IFN-}\gamma$
- Activate **NF- κ B** (TF)
- NF- κ B master-switch of the immune system
- $\text{TNF-}\alpha$
 - Prothrombotic
 - Leukocyte adhesion & migration
 - Macrophage activation
 - Hematopoiesis and lymphocyte development



Second line: Inflammation

The four major plasma enzyme systems:

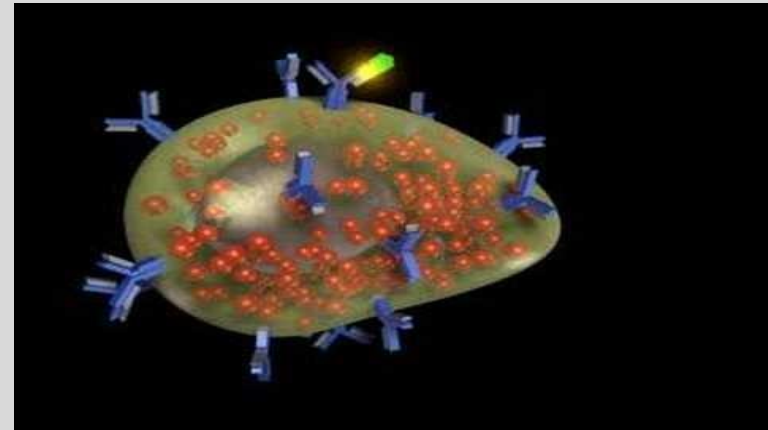
1. Clotting system
2. Fibrinolytic (plasmin) system
3. Kinin system
4. Complement system

Second line: Inflammation

- The plasmin system (*tissue repair & remodeling*):
 - a) Dissolve fibrin
 - b) Activates matrix metalloproteases (remodeling of collagen)
 - c) Promote angiogenesis
- The Kinin system (*bradykinin*)
 - a) Venular dilation due to release of nitric oxide (NO)
 - b) Increased vascular permeability
 - c) Smooth muscle contraction

Second line: Inflammation

- Other mediators:
 - Mast cells, basophils & platelets release histamine & serotonin (vasodilation & increased vascular permeability) & pain
 - IgE sensitizes mast cells & basophils
 - C3a/C5a trigger mast cell degranulation
 - Leukotrienes & prostaglandins



<https://www.youtube.com/watch?v=eVBqMXMIFnM>

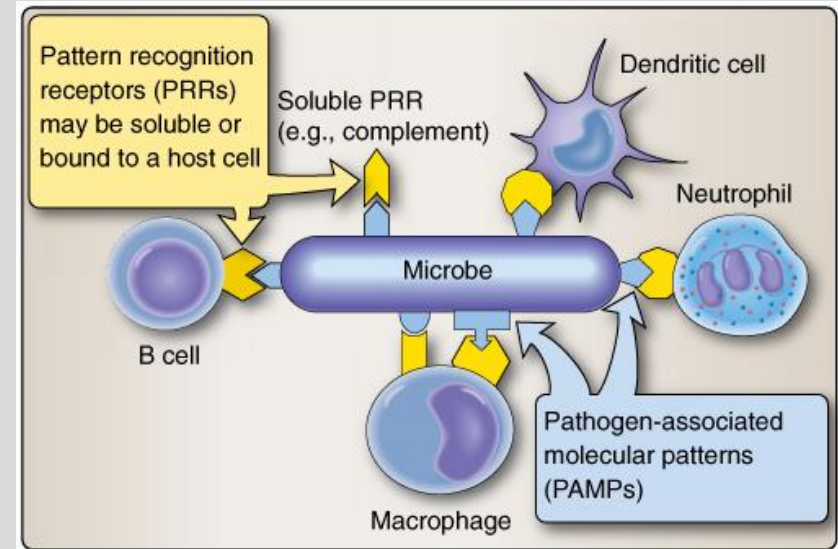
Second Line: Inflammation

mediator	main source	actions
histamine	mast cells, basophils	increased vascular permeability, smooth muscle contraction, chemokinesis
5-hydroxytryptamine (5HT – serotonin)	platelets, mast cells (rodent)	increased vascular permeability, smooth muscle contraction
platelet activating factor (PAF)	basophils, neutrophils, macrophages	mediator release from platelets, increased vascular permeability, smooth muscle contraction, neutrophil activation
IL-8 (CXCL8)	mast cells, endothelium, monocytes and lymphocytes	polymorph and monocyte localization
C3a	complement C3	mast cell degranulation, smooth muscle contraction
C5a	complement C5	mast cell degranulation, neutrophil and macrophage chemotaxis, neutrophil activation, smooth muscle contraction, increased capillary permeability
bradykinin	kinin system (kininogen)	vasodilation, smooth muscle contraction, increased capillary permeability, pain
fibrinopeptides and fibrin breakdown products	clotting system	increased vascular permeability, neutrophil and macrophage chemotaxis
prostaglandin E ₂ (PGE ₂)	cyclo-oxygenase pathway, mast cells	vasodilation, potentiates increased vascular permeability produced by histamine and bradykinin
leukotriene B ₄ (LTB ₄)	lipoxygenase pathway, mast cells	neutrophil chemotaxis, synergizes with PGE ₂ in increasing vascular permeability
leukotriene D ₄ (LTD ₄)	lipoxygenase pathway	smooth muscle contraction, increasing vascular permeability

David Male, Jonathan Brostoff, David Roth, and Ivan Roitt. 2013. *Immunology* 8th edition. Elsevier publisher

Recognition

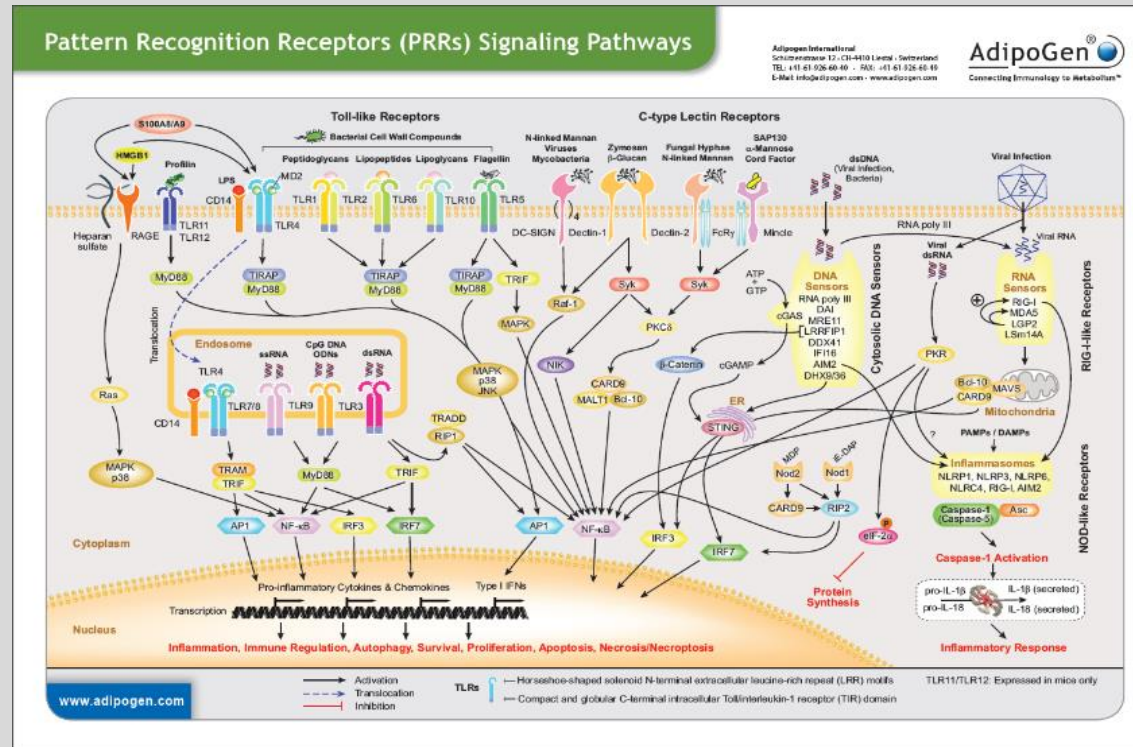
- Pattern recognition receptors (PRR)
 1. **Secreted** molecules (serum and bodily fluids; e.g., CRP, MBL)
 2. **Receptors** present on the cell surface & on endocytic vesicles (TLR)
 3. **Intra-cytoplasmic** recognition molecules (NOD-like receptors)
- Bind to PAMPs (LPS, PG)
- Coded by genome/no rearrangement/identical in all individuals



Source: Thao Doan, Roger Melvold, Susan Viselli and Carl Waltenbaugh. 2013. Lippincott's Illustrated Reviews: Immunology, 2ed. Lippincott Williams & Wilkins publisher

PRRs

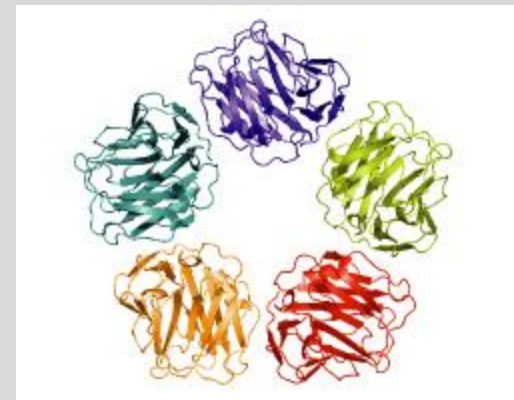
- Trigger inflammation
- Immobilizes the infectious microbe
- Promote ingestion by phagocytes
- Activate host cell secretion of antimicrobial substances



http://www.adipogen.com/media/Catalogs/Pix/Pattern_Recognition_Wallchart_Pix.PNG

PRRs (CRP)

- CRP: main acute-phase protein produced by hepatocytes in response to IL-6
- Member of the “pentraxins” family of proteins
- Opsonizes microbes (pneumococci)
- Activate classical complement pathway
- Binds to necrotic/apoptotic cells



Source: https://resources.rndsystems.com/images/site/c_reactive5727.jpg

Which of the following is a pattern recognition receptor (PRR)?

LPS

0%

Peptidoglycan

0%

CRP

0%

IL-2 R

0%

TLRs

- Found in most tissues (10 TLR)
- Activate phagocytes → APCs
- Activate NF-κB
- TLR3 deficiency susceptibility to HSV
- TLR7 higher viral load & disease progression in HIV

Receptor	Location	Ligand	Pathogen
TLR1	Cell surface	Lipopeptides	Gram -negative bacteria mycobacteria
TLR1/TLR2	Cell surface	Tri-acyl lipoprotein	Bacteria
TLR2	Cell surface	Lipoteichoic acid Lipoarabinomannan Zymosan Glycoinositol-phospholipids	Gram-positive bacteria Mycobacteria Fungi T. Cruzi
TLR2/TLR6	Cell surface	Di-acyl lipoprotein	Bacteria
TLR3	Endosome or Cell surface	dsRNA	Viruses
TLR4	Cell surface	LPS	Gram -negative bacteria
TLR5	Cell surface	flagellin	Bacteria
TLR6	Cell surface	Di-acyl lipopeptides	Mycobacteria
TLR7	Endosome	ssRNA	Viruses
TLR8	Endosome	ssRNA	Viruses
TLR9	Endosome	Unmethylated CpG DNA	Bacteria

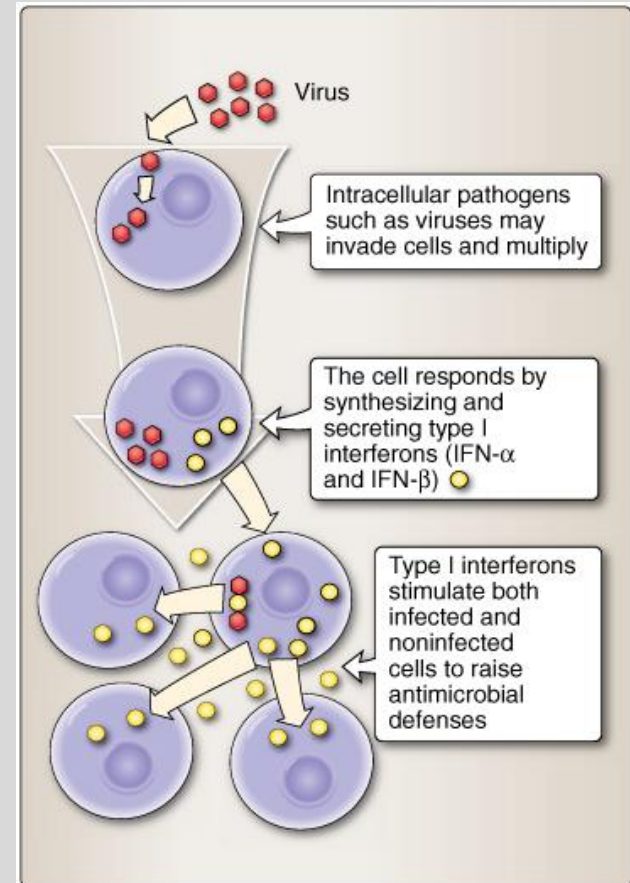
David Male, Jonathan Brostoff, David Roth, and Ivan Roitt. 2013. *Immunology* 8th edition. Elsevier publisher

Second line: Fever

- Pyrogens exogenous (LPS) or endogenous (e.g., IL-1 & IL-6)
- Benefits:
 - 1) Inhibits multiplication of temperature-sensitive microorganisms
 - 2) Impedes the nutrition of bacteria by reducing the availability of iron
 - 3) Increases metabolism & stimulates immune reactions & naturally protective physiological processes

Second line: Interferon

- IFN- α (DCs) & IFN- β (fibroblasts)
- RNA-dependent protein kinase $\rightarrow$ blocks mRNA synthesis and activate apoptosis
- IFN- γ (adaptive IR)



Source: Thao Doan, Roger Melvold, Susan Viselli and Carl Waltenbaugh. 2013. Lippincott's Illustrated Reviews: Immunology, 2ed. Lippincott Williams & Wilkins publisher

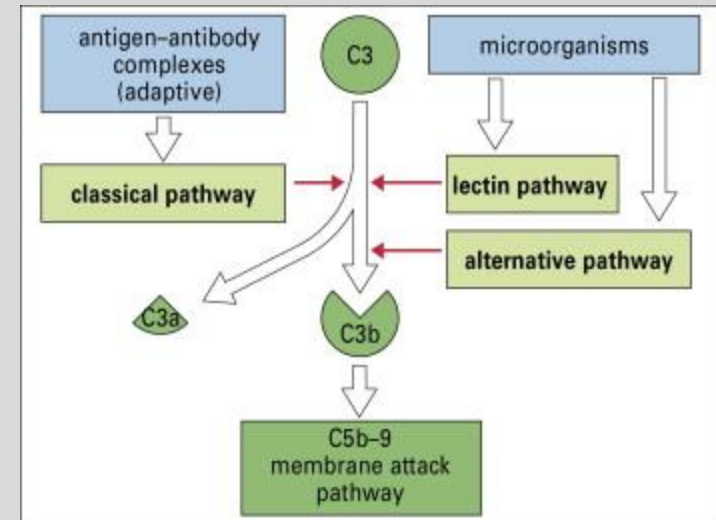
Second line: Complement

~16 soluble serum proteins
(10%)

Inactive; ancient

Activated

- Classical pathway (adaptive IR)
- Alternative pathway (innate IR)
- Mannan-binding lectin (MBL) pathway (innate IR)

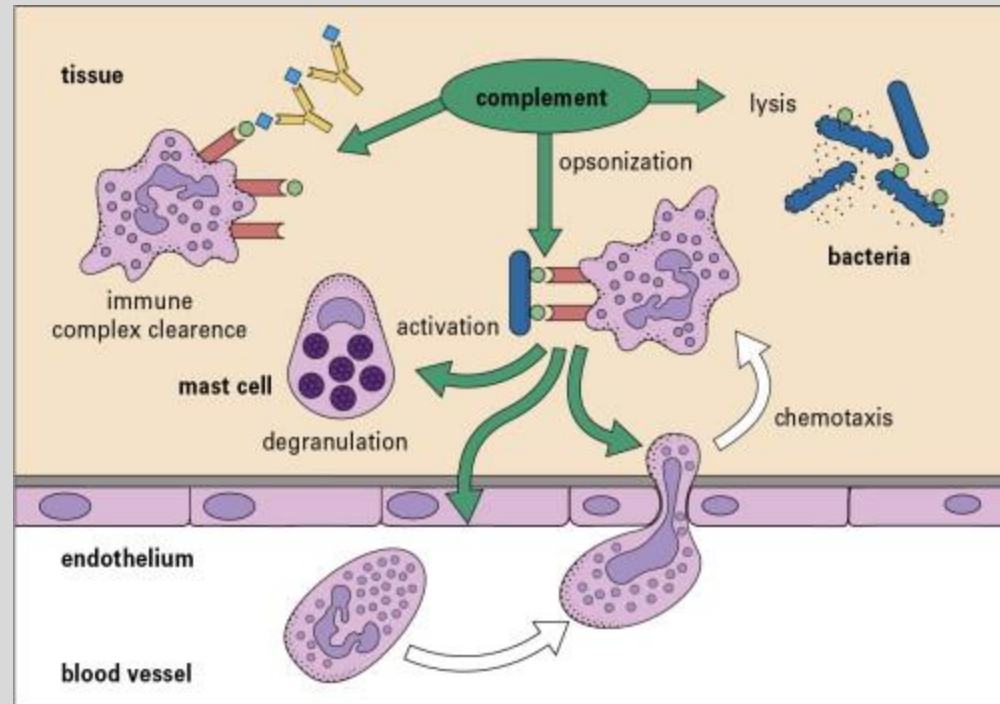


David Male, Jonathan Brostoff, David Roth, and Ivan Roitt. 2013. *Immunology* 8th edition. Elsevier publisher.

Second line: Complement function

- Function:

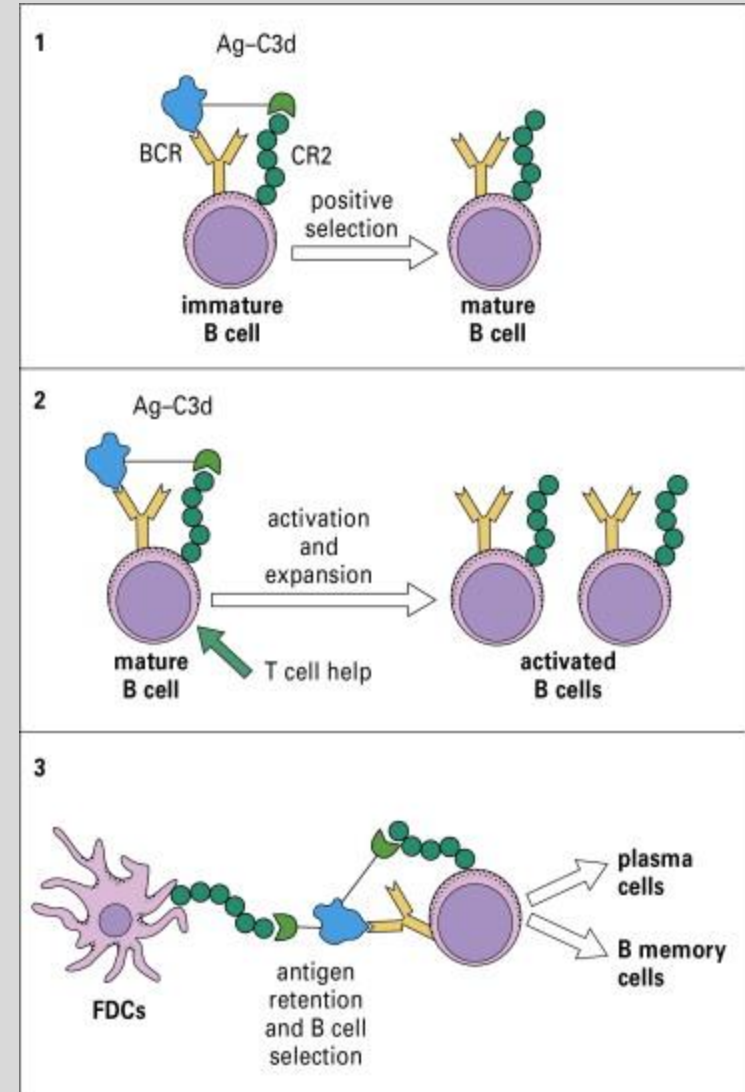
1. **Opsonization** (*C3b* & *C4b*)
2. Lysis of bacteria and enveloped viruses (*MAC*)
3. Inflammation & chemotaxis (*C3a* & *C5a*)
4. Efficient development of Ab response (*priming of B-cells*)
5. Clearance of immune complexes and apoptotic cells



David Male, Jonathan Brostoff, David Roth, and Ivan Roitt. 2013. *Immunology* 8th edition. Elsevier publisher

Second line: Complement C3b

- C-depleted & C-deficient mice have markedly reduced Ab responses
- C-opsonized particles are 1000x > active than un-opsonized in triggering Ab production
- C3b triggers:
 - B-cell maturation & migration to lymphoid organs
 - B-cell activation by binding to opsonized Ag
 - DC select for correct B-cell



Second line: Complement (activation)

- Classical: Ab (IgM/IgG) bound to antigen and requires Ca^{++}
- Lectin: microbial carbohydrates (mannose & NAG)
- Alternative:
 - Constant state of low-level activation
 - Requires Mg^{++} & factor B
 - Activated by surface molecules (i.e., LPS)

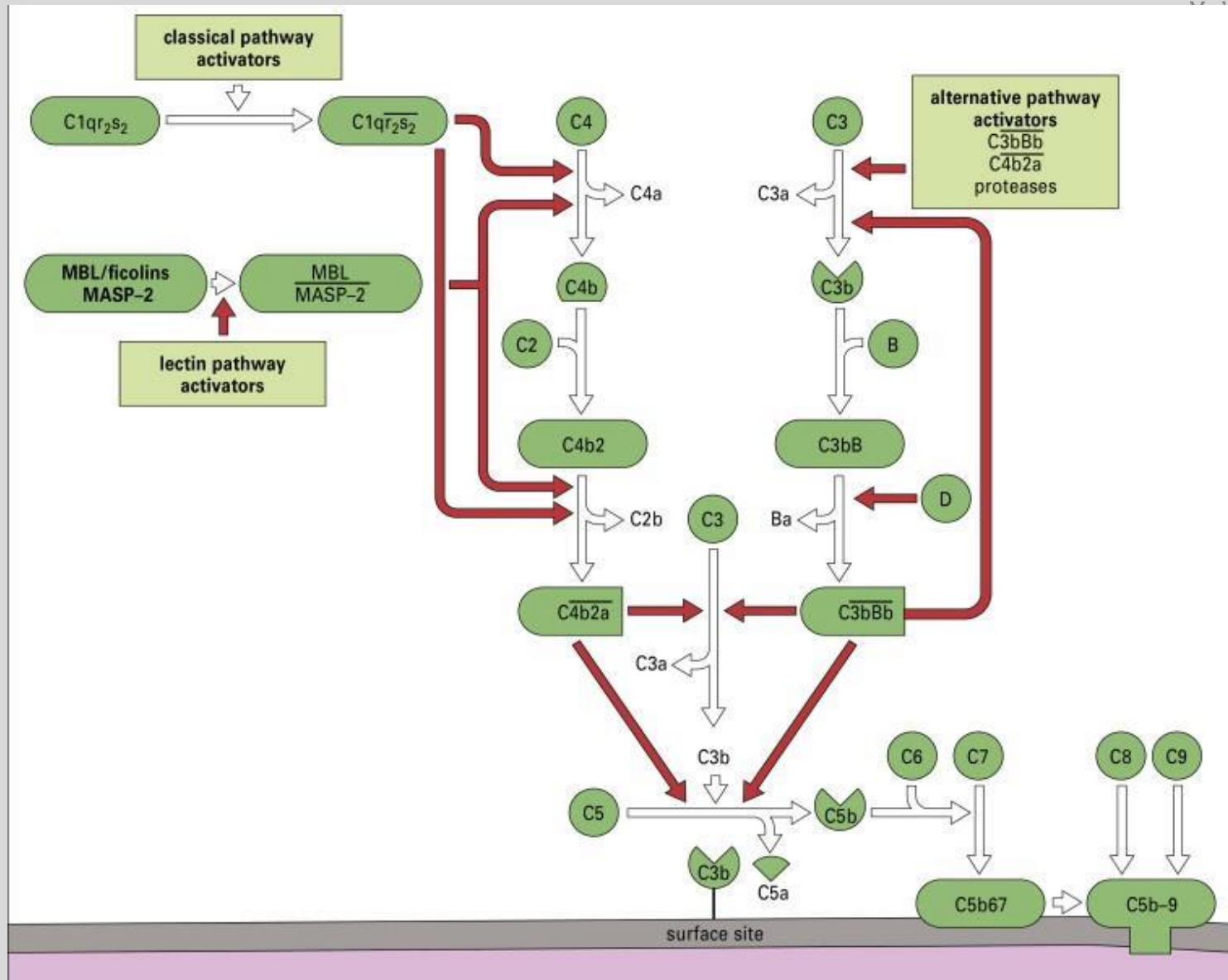
Second line: Complement

- Cascade amplification

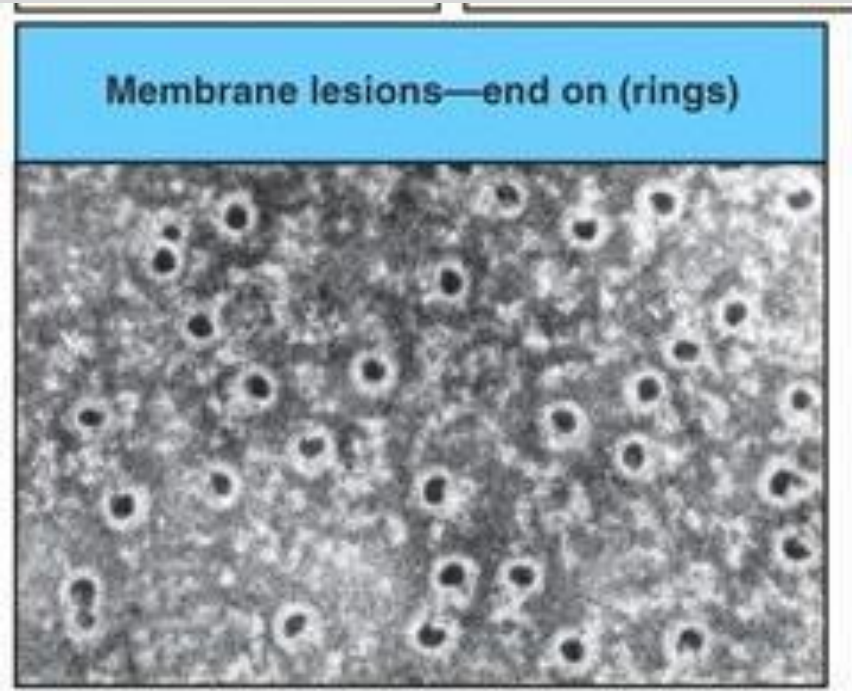
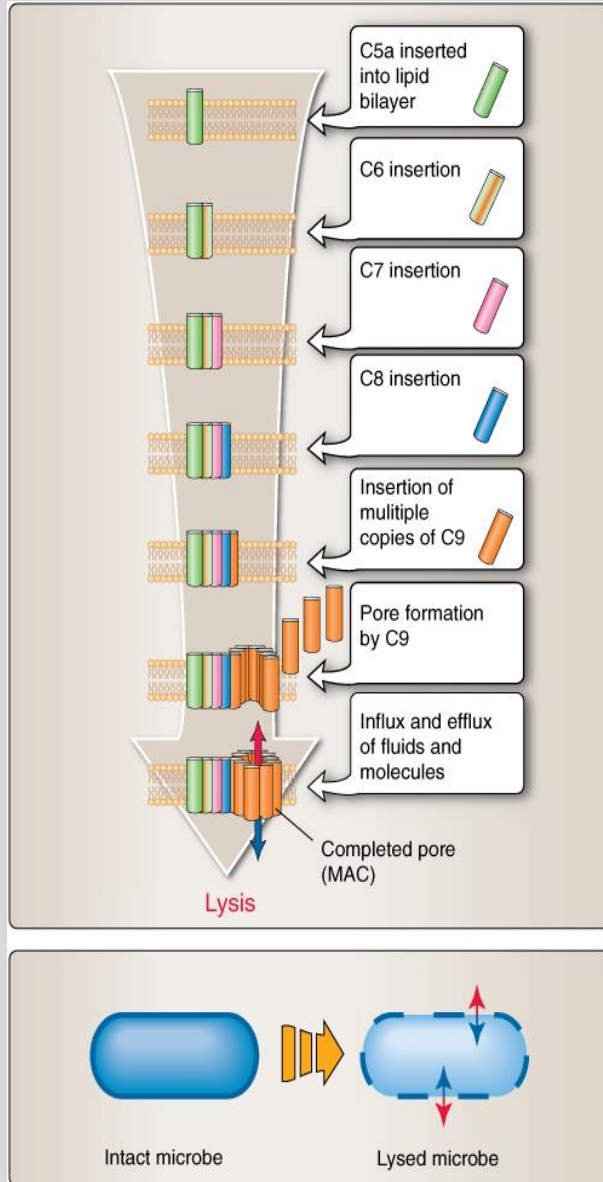
	immunoglobulins	microorganisms			other
		viruses	bacteria	other	
classical pathway	immune complexes containing IgM, IgG1, IgG2, or IgG3	HIV and other retroviruses, vesicular stomatitis virus		<i>Mycoplasma</i> spp.	polyanions, especially when bound to cations PO_4^{3-} (DNA, lipid A, cardiolipin) SO_4^{2-} (dextran sulfate, heparin, chondroitin sulfate)
lectin pathway		HIV and other retroviruses	many Gram-positive and Gram-negative organisms		arrays of terminal mannose groups acetylated sugars
alternative pathway	immune complexes containing IgG, IgA, or IgE (less efficient than the classical pathway)	some virus-infected cells (e.g. by Epstein-Barr virus)	many Gram-positive and Gram-negative organisms	trypanosomes, <i>Leishmania</i> spp., many fungi	dextran sulfate, heterologous erythrocytes, complex carbohydrates (e.g. zymosan)

David Male, Jonathan Brostoff, David Roth, and Ivan Roitt. 2013. *Immunology* 8th edition. Elsevier publisher

Second line: Complement

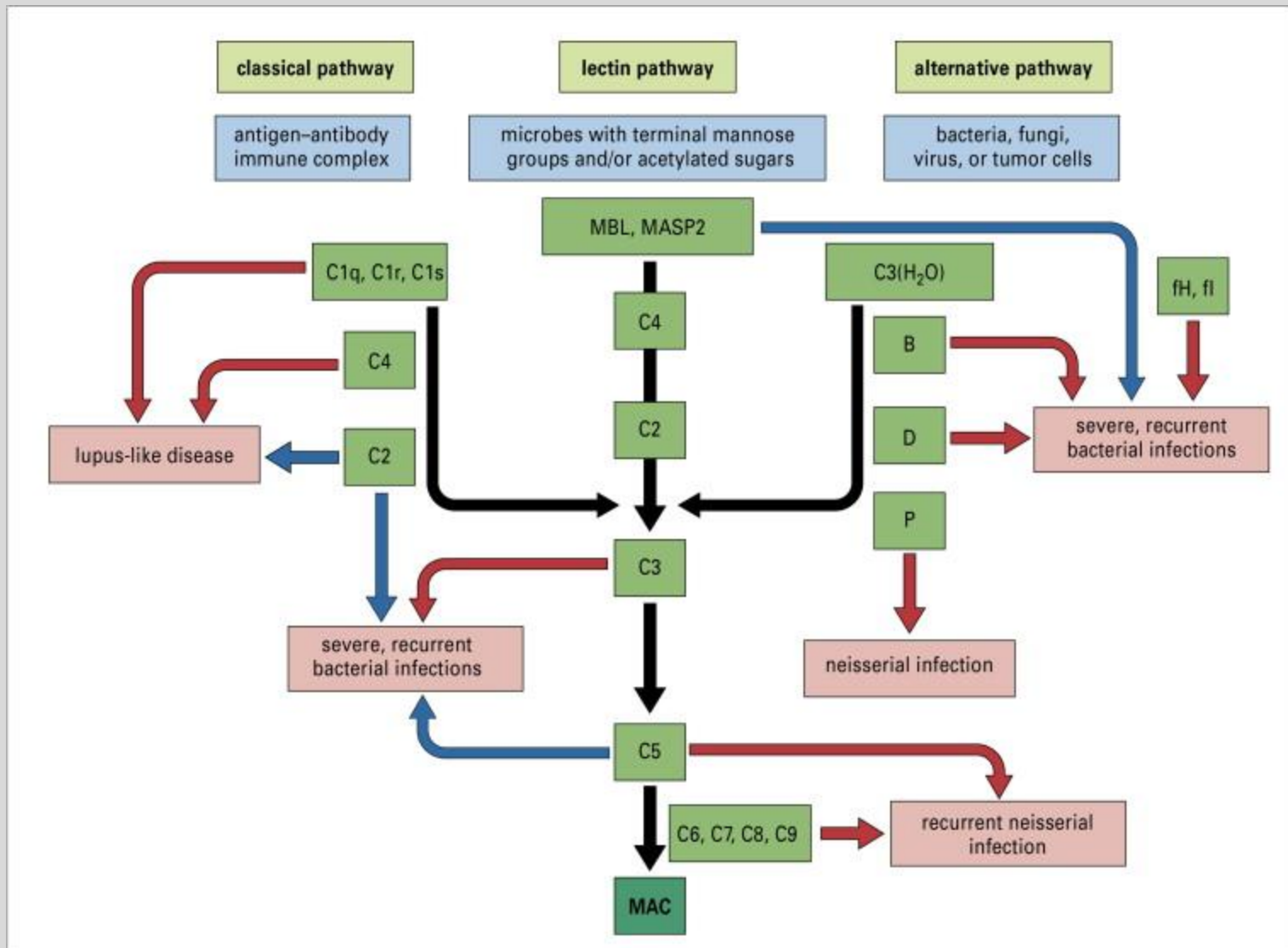


Second line: Complement (MAC)



Source: <https://s-media-cache-ak0.pinimg.com/originals/c8/d1/93/c8d19365c4cb369983eb20e4aa6e3b6c.jpg>

Second Line: Complement deficiencies



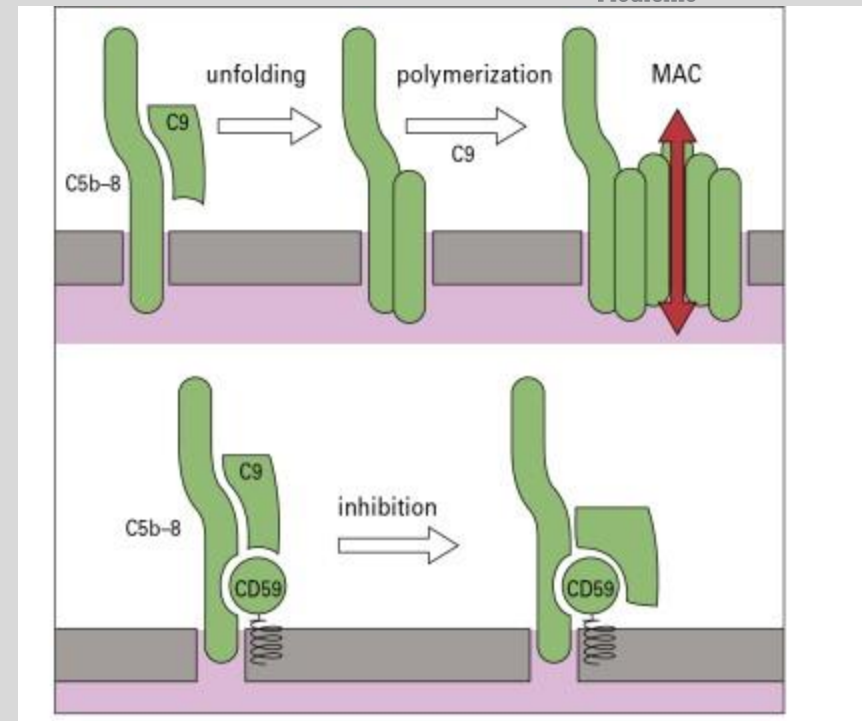
Second Line: Complement

- Complement protection systems:
 - a) Prevent consumption of components
 - b) Protect host from damage

	number of SCR domains*	dissociation of C3 and C5 convertases		cofactor for factor I on		localization
		classical pathway	alternative pathway	C4b	C3b	
C4b binding protein (C4bp)	52 or 56 in 7 or 8 chains	+	–	+	–	plasma
factor H (fH)	20	–	+	–	+	plasma
decay accelerating factor (DAF) (CD55)	4	+	+	–	–	on most cells including blood cells, endothelia, and epithelia
membrane cofactor protein (MCP) (CD46)	4	–	–	+	+	on most cells including blood cells (but not erythrocytes), endothelia, and epithelia
complement receptor 1 (CR1) (CD35)	30	+	+	+	+	erythrocytes, B cells, follicular dendritic cells, macrophages
*SCR, short consensus repeat						

Second Line: Complement (CD59)

- Inhibits binding of C9
- Is broadly expressed
- CD59 deficiency:
Paroxysmal nocturnal
hemoglobinuria (PNH)



David Male, Jonathan Brostoff, David Roth, and Ivan Roitt. 2013. *Immunology* 8th edition. Elsevier publisher



Source: <http://emedicine.medscape.com/article/207468-overview>

Which antibody is the most efficient activator of the classical pathway of complement?

0%

IgM

0%

IgG

0%

IgA

0%

IgE

0%

IgD

A 5-year-old male is born with a defect in the C3 complement protein. This will likely result in?

0%

Susceptibility to
bacterial
infections

0%

Susceptibility to
viral infections

0%

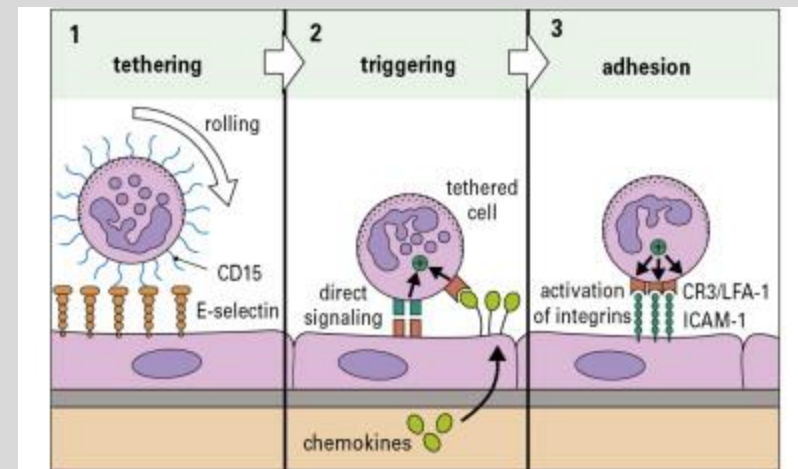
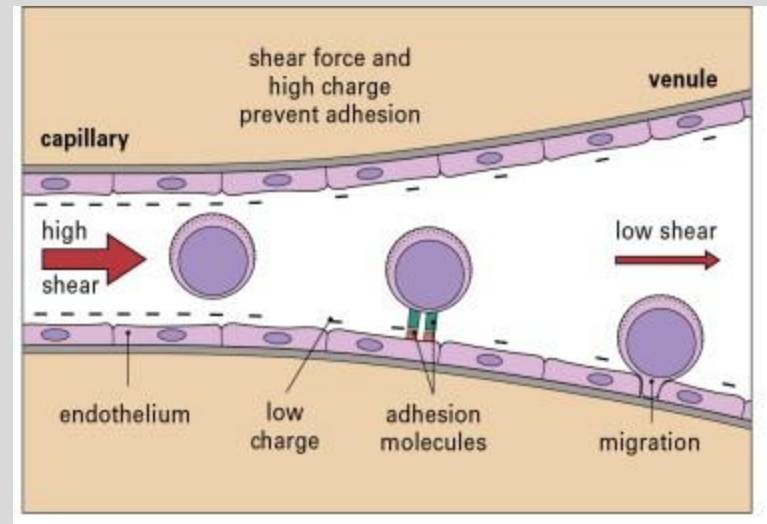
Increase in
inflammatory
response

0%

Increase in
autoimmune
disorders

Second line: Phagocytosis chemotaxis

- Respond to chemokines (> 50)
- Selectins & Integrins
- Adhesion to selectively-expressed *selectin* receptors in venules
- Integrins bind to CAMs (ICAM & VCAM)



Treatment of a patient with monoclonal antibodies against "integrins" would likely result in?

Prevention of phagosome-lysosome fusion

0%

Inhibition of intracellular pathogen recognition

0%

Increase susceptibility to type I hypersensitivity reactions

0%

Inhibition of migration of phagocytes

0%

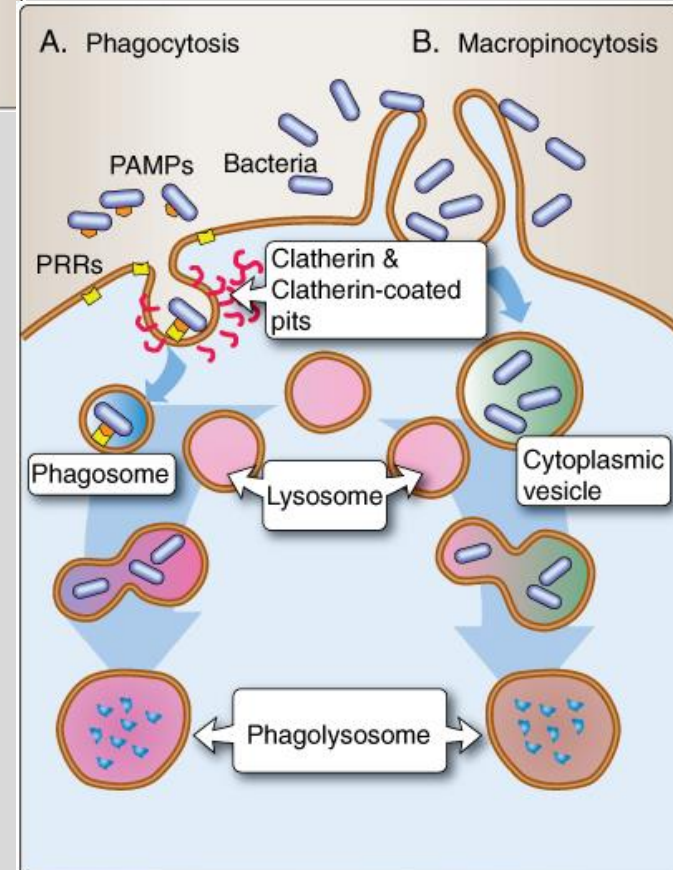
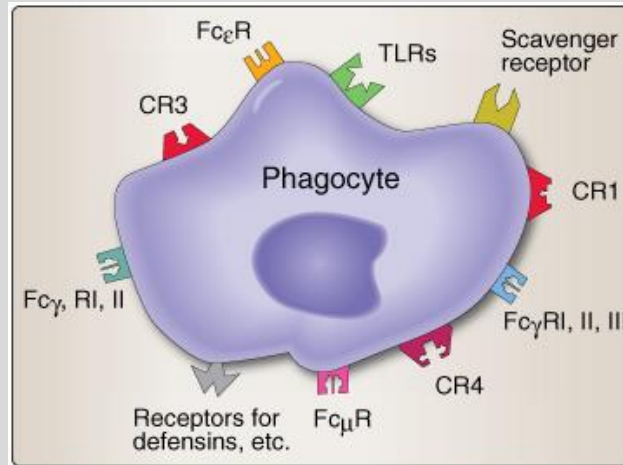
Chemokines & their receptors

- Both ligand/receptors vary resulting in selection of leukocytes entering tissues
- Defect: leukocyte adhesion deficiency (LAD) syndrome (β -integrins)
- Therapeutic application: Abs against VLA-4 (the ligand for VCAM-1) MS patients (limit migration of T-cells into the CNS)

	Th1		MØ		Th2		eosinophil	basophil	neutrophil
	(CCR1)	CCR1	CCR2	CCR2	CCR5 → CCR5	CCR3 → CCR3	CCR3 → CCR3	CCR3 → CCR3	(CCR1)
	CCR2	CCR2	CCR5	CCR5	CCR3	CCR3	CCR3	CCR3	CXCR1
	CXCR3								CXCR2
CCL3	+	+				+	+		+
CCL4	+	+							
CCL5	+	+			+	+	+		+
CCL2	+	+			+	+	+		
CCL7,8	+	+			+	+	+		+
CCL13	+	+			+	+	+		
CCL11					+	+	+		
CXCL8									+
CXCL1,2,3									+
CXCL9,10,11	+								

Second line: Phagocytosis (Process)

- Attachment (TLR/CR/Fc)
- Recognition ('don't eat me' signal CD47)
- Engulfment (*phagosome*)
- Fusion of phagosome with lysosome
- Digestion
- Expulsion/Ag presentation



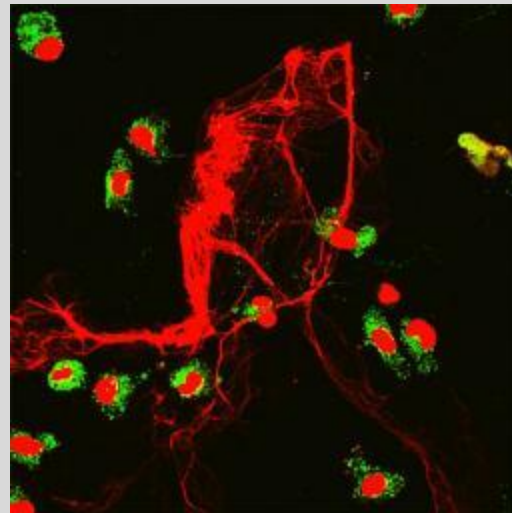
Source: Thao Doan, Roger Melvold, Susan Viselli and Carl Waltenbaugh. 2013. Lippincott's Illustrated Reviews: Immunology, 2ed. Lippincott Williams & Wilkins publisher

Second Line: Phagocytes (intracellular receptors)

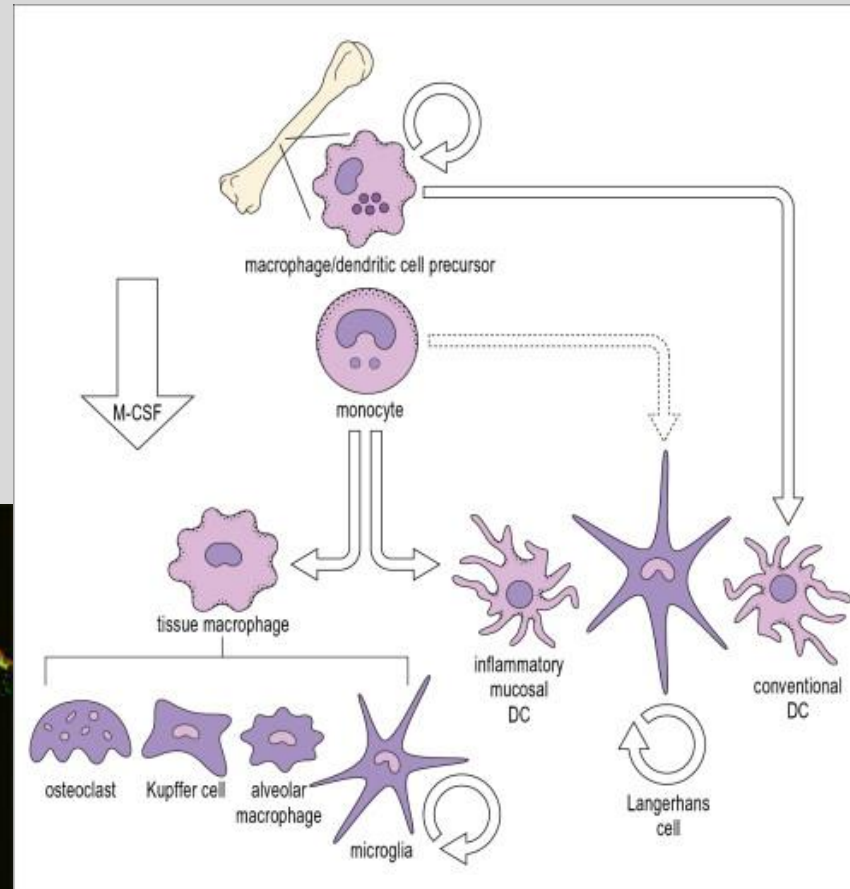
- Cytosolic receptors recognize intracellular pathogens (part of PRRs)
 - a) NLR (*NOD-like receptors*) - PG
 - b) RLHs (*Retinoic acid-inducible gene I-like helicases*) - dsRNA
- NLRs form the "**inflammasome**" triggers inflammatory cell death (pyroptosis) of infected cells
- NOD polymorphisms:
 1. Atopic eczema
 2. Asthma
 3. Increase IgE
 4. Increase risk of Crohn's disease
 5. Increase in bacterial infections

Second line: Phagocytes (Types)

- PMNs: Neutrophils (recruited by Macrophages) & eosinophils
- MN cells:
Monocytes/Macrophages, DC
& osteoclasts



<http://people.eku.edu/ritchisong/301images/Eosinophil.jpg>



David Male, Jonathan Brostoff, David Roth, and Ivan Roitt. 2013. *Immunology* 8th edition. Elsevier publisher

Second line: Phagocytes (MΦ)

Macrophage populations

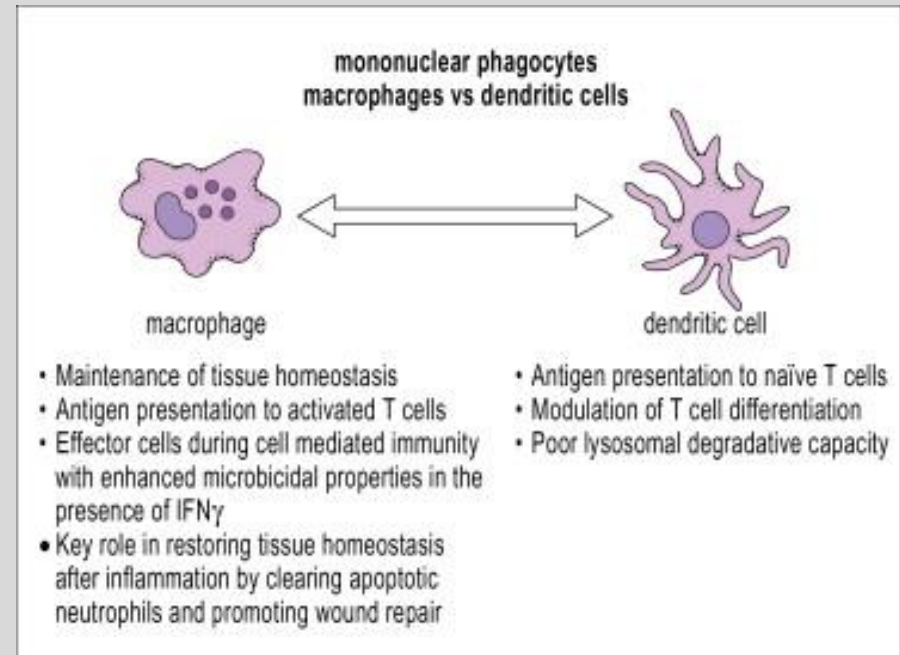
organ	name/site	functions/properties
bone marrow	stromal macrophage	interacts with hematopoietic cells, removes erythroid nuclei
liver	Kupffer cells	clearance of cells and complexes from the blood
spleen	red pulp macrophages white pulp tingible body macrophages marginal zone macrophages	clearance of senescent blood cells phagocytosis of apoptotic B cells interface between circulation and immune system
lymph node	subcapsular sinus macrophages medullary macrophages	interface with afferent lymph interface with efferent lymph
thymus	thymic macrophage	clearance of apoptotic cells
gut	lamina propria	endocytosis
lung	alveolar macrophage	clearance of particulates and infectious agents
brain	microglia in neuropil choroid plexus	interacts with neurons interface with cerebrospinal fluid
skin	Langerhans' cells	antigen capture
reproductive tract	ovary, testis	clearance of dying cells
endocrine organs	adrenal, thyroid, pancreas, etc.	metabolic homeostasis
bone	osteoclasts	bone remodeling

Second Line: Phagocytes (Neutrophils vs Macrophages)

- Key differences between Neutrophils & Macrophages
 - a) Neutrophils:
 - i. Short lived
 - ii. Acute response
 - iii. Fast clearance
 - iv. Store a wide range of antimicrobial polypeptides in intracellular granules (~50% defensins)
 - v. Readily produce oxygen radicals
 - b) MΦ:
 - i. Long lived
 - ii. Chronic inflammation/infection
 - iii. Less microbicidal and cytotoxic
 - iv. Recruit neutrophils
 - v. Requires activation by PRRs
 - vi. Display a higher degree of specificity which makes them better suited as sentinels of the immune system

Second line: Phagocytes' cytokines

- Secrete:
 - a) IL-1 (pyrogen)
 - b) TNF- α (inflammation)
 - c) IL-8 & IL-12 (attract neutrophils & NK cells)



Second line: Phagocytosis (evasion)

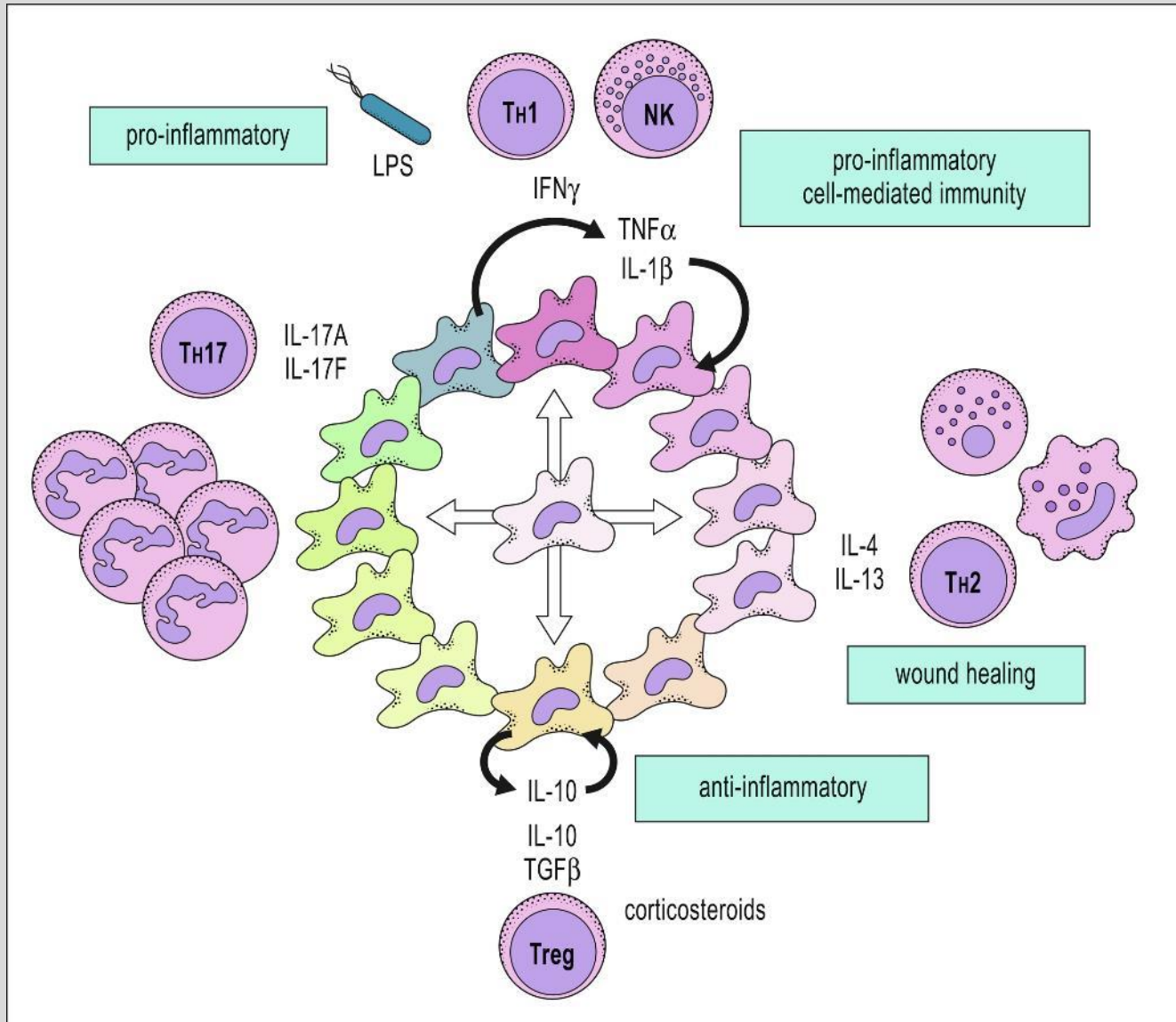
- Pathogens evade phagocytosis:
 - a) Capsule
 - b) Inhibit phagolysosome fusion (*M. tuberculosis*)
 - c) Escape phagosome (*L. monocytogenes*)
- Infectious agents can stimulate autophagy in Macrophages (*M. tuberculosis*, *T. gondii*, *F. tularensis*, & *Salmonella* species)

Second line: Phagocytes(MΦ types)

MΦ phenotypes (M1/M2)

Th1 type		Th2 types		significance
IFN γ		IL-4/IL-13	IL-10	
MHC class II molecule (Ia)	++	+	–	immune cell interactions
respiratory burst NO	++ ++	(–) (–)	– –	cell-mediated immunity tissue injury (e.g. tuberculosis)
TNF α IL-1 IL-6	++	(–)	–	proinflammatory
mannose receptor	–	++	+	phagocytosis/endocytosis (e.g. antigens)
growth	–	++	0	local growth of MØ in immune lesions
fusion	0	++	0	giant cell formation–granulomas
growth factor secretion	–	++	+	healing of lesions

Second line: Phagocytes(MΦ regulation)



Monoclonal antibodies targeting CD47 molecules on cells would likely result in?

Phagocytosis of the cells

0%

Inhibition of phagocyte recognition

0%

Increase in type I hypersensitivity reactions

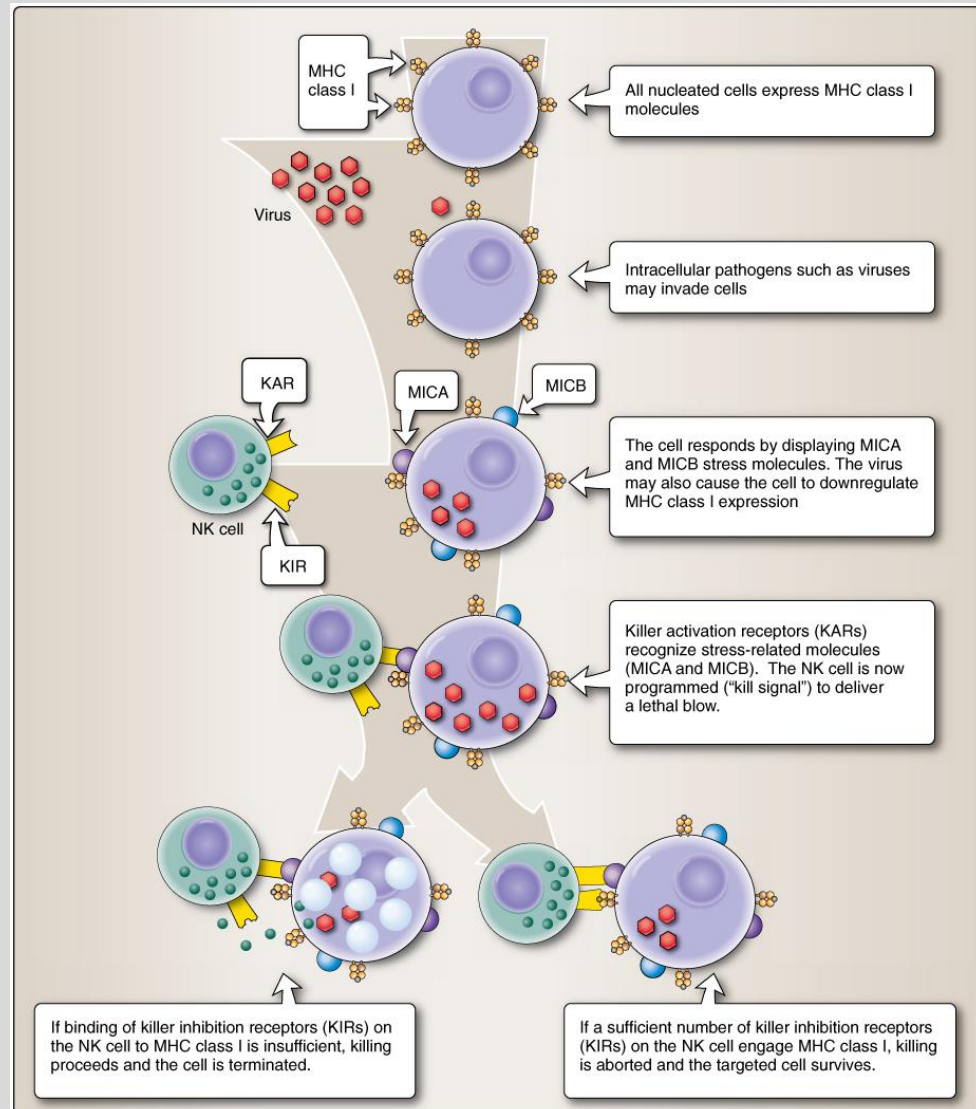
0%

Serum sickness

0%

Second line: NK cells

- NK cells have killer activation (KAR) & killer inhibition receptors (KIR)
- KAR recognize stress-associated molecules (MICA, MICB)
- NK cells use killer inhibition receptors (KIR) to assess MHC I (insufficient levels of KIR-MHC I → killing target host cell)
- Viruses & malignant events depress MHC I



Primary Immunodeficiencies in innate IR

Phagocyte disorders:

1. *Chediak-Higashi syndrome*: defect in phagosome-lysosome fusion
2. *Chronic Granulomatous disease*: defect in microbicidal enzymes/O₂ radicals
3. *IFN- γ receptor* deficiency
4. *Leukocyte adhesion deficiency* (LAD-1/LAD-2)

Phagocytes were isolated in the lab from a mouse species that are highly susceptible to infections. In vitro, the mouse phagocytes were able to form phagolysosomes and move in response to a chemokine but they weren't able to produce oxygen radicals, such as superoxide. These mice likely suffer from the human equivalent of?

Chronic granulomatous disease

0%

Chediak-Higashi syndrome

0%

Hereditary angioedema

0%

Leukocyte adhesion deficiency (LAD-1)

0%

Primary Immunodeficiencies in innate IR

Complement disorders:

1. Defects in **alternative** pathway & **MBL** lead to increase susceptibility to infections
2. Defects in **classical** pathway not associated with significant increase in susceptibility to infections (ex. C3 deficiency & *encapsulated*)
3. Deficiencies in C1, C2 & C4 → inefficient clearance of immune complexes (increase in type III hypersensitivity disease)
4. Deficiency in C regulation (C1 inhibitor = hereditary angioedema)

Summary Slide

- Compared innate IR to adaptive IR
- Classified innate IR into 1st and 2nd line of defense
- Discussed role of the 1st line of defense (physical and chemical)
- Discussed role of the 2nd line of defense (inflammation, complement, phagocytosis, etc.)
- Pointed out deficiencies in these systems and how that leads to diseases or syndromes

References

Core References:

1. David Male, Jonathan Brostoff, David Roth, and Ivan Roitt. 2021. **Immunology** 9th edition. Elsevier publisher. ISBN: 978-0-323-08058-3
2. Thao Doan, Roger Melvold, Susan Viselli and Carl Waltenbaugh. 2022. **Lippincott's Illustrated Reviews: Immunology**, 3rd edition. Lippincott Williams & Wilkins publisher. ISBN: 978-1-4511-0937-5
3. Tao, LE, V. Bhushan and M. Sochat. 2025. **First Aid for the USMLE Step 1**; McGraw Hill Publication.

Lecture Feedback Form:

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>